

POWER WINDOW SYSTEM

■ DESCRIPTION

- The power window regulator motor assembly of the new Land Cruiser contains an ECU and Hall IC. The ECU effects power window control, and the Hall IC detects the position and the direction of the motor.
- The power window system has the following functions:

Function	Outline
Manual up-and-down (All Doors)	This function causes the window to open or close while the power window regulator switch is being pulled halfway up or pushed halfway down. The window stops as soon as the switch is released.
One touch auto up-and-down (All Doors)	The one touch auto up-and-down function enables the window to be fully opened or closed with a single touch of the power window regulator switch.
Variable speed control	This control minimizes the occurrence of noise during the power window open/close operation by changing the operating speed of the power window motor.
Jam protection (All Doors)	The jam protection function automatically stops the power window and moves it downward if a foreign object gets jammed in the window during one touch auto up operation or manual up operation.
Remote control (All Doors)	The multiplex network master switch assembly can control the up-and-down operations of the windows.
Window lock	Power window operation of the 3 passenger windows is disabled when the window lock switch is pressed.
Key off operation	This function enables the power window to be operated within approximately 43 seconds of the engine switch being turned off. However, if either the driver door or front passenger door is opened within this 43 seconds, the power window operation will be disabled. If the one touch auto up-and-down function is currently operating when the 43 seconds have elapsed, the power window will continue to move until it is fully open or closed.
Transmitter linked up-and-down*	When a lock/unlock signal is transmitted by the key (electronic key) for longer than 2.4 seconds to the certification ECU, the certification ECU transmits the window close/open signal to the main body ECU. Power window ECUs activate all the power window motors in accordance with the signals from the main body ECU to open or close all the windows.
Lock switch linked up*	All the windows are closed when the lock switch on the outside door handle is pressed for 3 seconds or more with the key in the detection area and all the doors locked. This function can be performed using the door lock switch of any door.
Mechanical key linked up-and-down*	When the key (electronic key) is not in the interior detection area, the driver door is locked, and the key (mechanical key) in the driver door key cylinder is turned and maintained in the lock direction for 1.5 seconds or more, the main body ECU activates the power window motor to close all the door windows while the key (mechanical key) is turned. Similarly, when the driver door is unlocked, turning and maintaining the key (mechanical key) in the driver door key cylinder in the unlock direction for 1.5 seconds or more will cause all the door windows to be opened.
Diagnosis	When any power window ECU detects either of the following conditions, the self-diagnosis function switches the ECU to fail-safe mode. • An abnormality in the Hall IC that detects the position, speed, and direction of the window. • An error in the window detection position and the upper limit position recorded in the power window ECU.

(Continued)

Function	Outline
Fail-safe	If the Hall IC in the power window motor malfunctions, some power window functions will be prohibited by the fail-safe mode: Each power window operates when the corresponding power window regulator switch is fully pushed down or pulled up and held in that position.

*: These functions are only for VX grade models. The setting function can be changed using the customized body electronics system. For details, refer to Customized Body Electronics System section on page BE-17.

Service Tip

The power window regulator motor assembly stores the initial position of each door window. The memory is not cleared if battery terminals, fuses or power window motor connectors are disconnected. However, the power window regulator motor assembly needs to be initialized if it is replaced with a new one or it is reinstalled with the power window regulator sub-assembly. When necessary, perform the initialization as follows:

Initialization after Reinstallation of the Power Window Regulator Motor Assembly

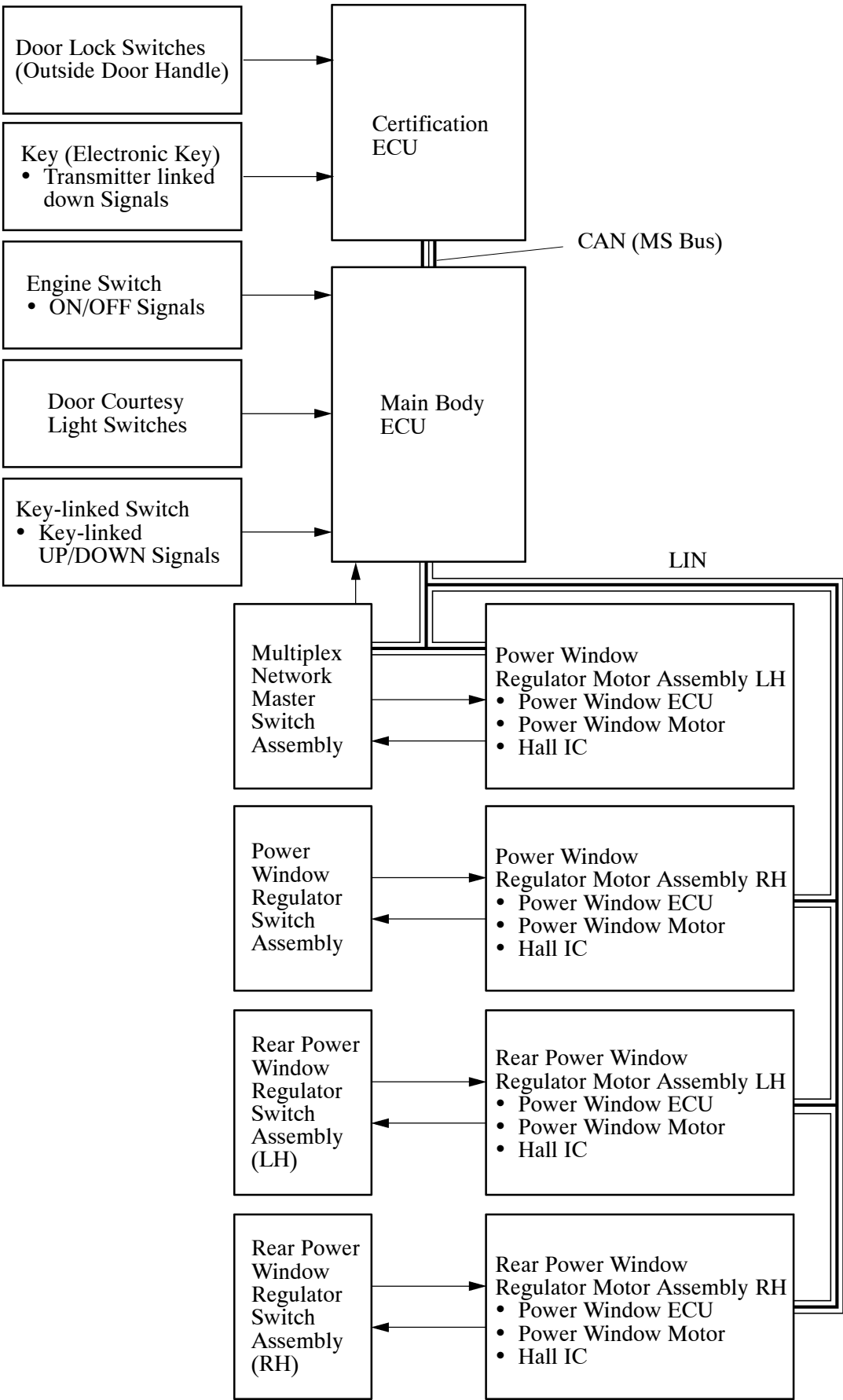
- Connect the battery and turn the engine switch on (IG).
- Pull up the power window regulator switch to the AUTO UP position and hold it until the window is fully closed.
- Hold the power window regulator switch in the AUTO UP position for at least 6 seconds after the window is fully closed.
- Push down the power window regulator switch until the window is fully opened.
- Hold the power window regulator switch for at least 1 second after the window is fully opened.
- Release the switch once, push down the power window regulator switch again and hold it for at least 4 seconds.
- Pull up the power window regulator switch to the AUTO UP position and hold it until the window is fully closed.
- Hold the power window regulator switch in the AUTO UP position for at least 1 second after the window is fully closed.
- Check that the power window regulator switch illumination has changed from flashing to being illuminated.
- Make sure that the window opens and closes automatically using the one touch function.

Initialization after Replacement of the Power Window Regulator Motor Assembly

- Connect the battery and turn the engine switch on (IG).
- Check that the power window regulator switch illumination blinks after the power source is turned on.
- Push down the power window regulator switch until the window is fully opened.
- Hold the power window regulator switch for at least 1 second after the window is fully opened.
- Pull up the power window switch to the AUTO UP position and hold it until the window is fully closed.
- Hold the power window regulator switch in the AUTO UP position for at least 1 second after the window is fully closed.
- Check that the power window regulator switch illumination has changed from flashing to being illuminated.
- Make sure that the window opens and closes automatically using the one touch function.

For details, see the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

► System Diagram ◀



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